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LETTER

Organic Hole Transport Material Ionization Potential Dictates Diffusion Kinetics of Iodine Species in Halide Perovskite Devices

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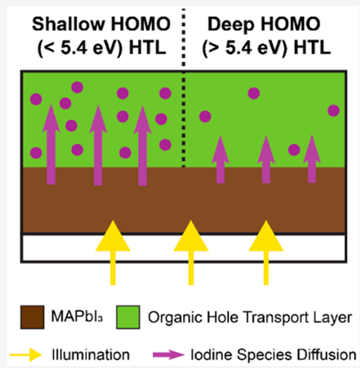
ABSTRACT: **Iodine-containing volatiles are major degradation products of halide perovskite materials under irradiation, yet iodine diffusion kinetics into and throughout organic hole transport materials (HTMs) and consequent reactions are largely unexplored. Here, we modify the Ca:O_2 corrosion test to Ag:I_2 to quantify I_2 transmission rates through common organic HTMs. We observe I_2 permeability to inversely correlate with HTM ionization energy, or the highest occupied molecular orbital (HOMO) energy. Tracking electronic conductance during exposure to I_2 confirms shallow HOMO HTMs are strongly oxidized (i.e., doped), leading to substantial I_2 uptake and increased transmission rates. Finally, relationships between HOMO level, doping, and transmission rate are maintained when methylammonium lead triiodide ($MAPbI_3$) photolysis products are the only source of iodine. While HTM energetics influence the initial performance of halide perovskite devices by selective charge extraction, our results further suggest they will affect device stability; deeper HOMO energy HTMs will suppress iodine migration and associated degradation mechanisms.**



Halide perovskite materials degrade under individual or combined stressors including visible light illumination, elevated temperature, electron beams, and X-ray irradiation, inducing substantial release of HI and molecular I_2.[1−6] These species are extremely corrosive and known to compromise device performance and stability due to iodization of the metal electrode.[7−12] The metal can be protected by a buffer layer that is less permeable to iodide/iodine (e.g., MoO_3/Al or Cr_2O_3/Cr),[9−12] but the details of iodide/iodine permeation throughout the device should still be considered.[13−18] High solubility/permeability of and doping by I_2 in organic semiconductors[19−22] is often overlooked, leading to perovskite device models that focus on ion accumulation at interfaces.[15,23−25] Interface accumulation incompletely describes the ensemble of microscopic processes occurring, the most salient being electric field-independent diffusion of net-neutral charged iodide/hole or triiodide/hole pairs within the organic hole transport material (HTM). Halide perovskite technologies can benefit from measurement techniques that characterize iodide/iodine diffusion rates as well as elucidate ionic speciation and reaction mechanisms influencing mass transport kinetics.

In this study, we adapt the electrical calcium corrosion test by replacing the Ca thin film with Ag, enabling us to quantify mass transport of iodine through HTMs commonly used in halide perovskite devices.[26] The Ag resistor is sensitive to corrosion by I_2 but unreactive toward H_2O and O_2, allowing measurements in atmosphere. By varying the device configuration, we examine (1) I_2 vapor transmission rates and permeability constants, (2) oxidation/doping by I_2 vapor exposure, and (3) doping by iodine-containing species sourced from *in situ* perovskite degradation.

Generally, and independent of molecular or polymeric structure, we observe HTMs with low ionization energy (<5.4 eV, referenced to vacuum), dictated by the highest occupied molecular orbital (HOMO) energy, to more rapidly transmit I_2 due to a doping reaction that facilitates uptake of a large amount of triiodide. The doping reaction is much less favorable for deeper HOMO energies (>5.4 eV), resulting in lower concentrations and fluxes of I_2 species. This relationship holds when iodide/iodine species are released at a methylammonium lead triiodide ($MAPbI_3$)/HTM interface under

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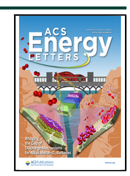

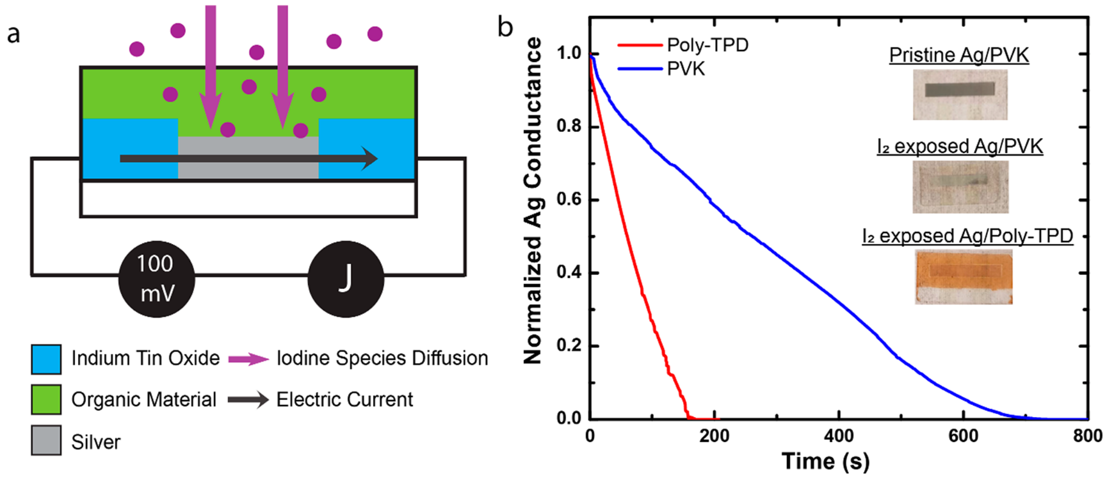

Figure 1. (a) Schematic of the Ag:I_2 corrosion test device for evaluating iodine diffusion rates through HTM thin films. (b) Conductance transients of the Ag resistor coated with poly-TPD or PVK after correcting for the ITO series resistance, and photographs of the devices before and after I_2 exposure. The Ag films are 2 cm long for scale.

UV illumination. Thus, while pinholes and grain boundaries remain viable routes for iodine transport through HTMs and must be mitigated, our data assert that HTM HOMO energy is an intrinsic parameter that determines the rate of transport through the HTM bulk. Critically, these results indicate that HTM HOMO energy may impact the stability of a halide perovskite device in addition to its initial performance; the HOMO should be well-matched to the perovskite valence band maximum for efficient hole collection, and should be as deep as is feasible to slow degradation caused by iodine diffusion.

I_2 Permeability Constant Measurement. First, we characterized I_2 vapor transmission rates through small-molecule and polymeric HTMs (HTM abbreviations, structures, and HOMO energies tabulated in the Supporting Information (SI), Table S1) using a modified Ca corrosion test.[26] The device structure, shown in Figure 1a, consists of a 30 nm thick Ag thin film resistor bridging a 5 mm wide channel between prepatterned indium tin oxide (ITO) electrodes. The HTMs were deposited on top of the Ag sensor by thermal evaporation or spin coating from solution. Transmission rates were measured by monitoring the current at 0.1 V bias as a function of time after exposure to I_2 vapor (partial pressure ∼0.2 Torr at STP). Representative transients in the Ag conductance are shown for poly(vinylcarbazole) (PVK) and poly(*N,N*′-bis(4-butylphenyl)-*N,N*′-bisphenylbenzidine) (poly-TPD) in Figure 1b. Correcting for the ITO series resistance (∼40 Ω) shows that the Ag (∼2−4 Ω) conductance falls linearly with time, suggesting steady-state conditions are established quasi-instantaneously (compared to the time scale of our measurements) and Fick's first law dictates the flux of I_2 species. Fickian behavior was observed for the polymers and small-molecule organics with shallow HTMs; deviations from Fick's laws were observed for *N,N*′-bis(napthalen-1-yl)-*N,N*′-bis-(phenyl)-benzidine (NPB) and 4,4′,4″-tris(carbazol-9-yl)-triphenylamine (TCTA) (SI, Figure S1) which may arise from morphological weak points dominating diffusion at early times since, as we will show, bulk transmission is lower for deep HOMO energy materials. This simple test proved useful to characterize factors that improve I_2 barrier properties of organic materials, such as densification of poly(methyl methacrylate) (PMMA) by annealing (SI, Figure S2), as well as use of organic HTM/MoO_3 composite bilayers (SI, Figure S3), making it generally attractive for understanding iodine-induced degradation.

Membrane diffusion model calculations suggest Fick's first law dominates measurements for the HTM thicknesses used here, which are relevant for perovskite devices.[27] The iodine flux, *F*, for our system is simply

$$F = \frac{(C_0 - C_L)D}{L} \tag{1}$$

where C_0 is the equilibrium concentration of iodine in the HTM just within the surface (assuming reaction with I_2 vapor is fast), C_L is the concentration at the HTM/Ag interface, which is assumed to be zero given a fast reaction between Ag and I_2 at room temperature (uncoated Ag sensors degraded in less than 10 s), D is the diffusion coefficient, and L is the HTM thickness. From eq 1, it is established that the measurement cannot deconvolute C_0D (extracted from the slope in Figure 1b). Further calculations (SI, Figure S4 and associated discussion) suggest that Fick's second law dynamics will display a "lag time" as the HTM takes up I_2 during the establishment of steady-state gradients; our calculations show that this may manifest for HTM thicknesses >5 μm if D and C_0 are of the order 10^{-9} cm²/s and 10^{20} atoms/cm³, respectively, which are reasonable upper limits of these parameters.[19] It may be possible to decouple C_0 from D for such a thick HTM barrier, but the properties likely differ substantially from the those of their thin-film counterparts.

C_0 and D cannot be quantified for our thin samples, so we use a permeability constant to describe the I_2 flux per eq 2:[27]

$$Q(t) = \frac{P\Delta pt}{L} \tag{2}$$

where $Q(t)$ is the total amount of I_2 that passes through the barrier, P is the permeability constant, t is time, L is the HTM thickness, and the pressure differential, Δp, is taken to be 0.2 Torr (the STP partial pressure of I_2). P can be easily obtained since $Q(t_{final})$ is equal to the number of moles of Ag per cm², determined by the Ag thickness and assumed density of 10.49 g/cm³.

Ag:I_2 corrosion tests were carried out on a number of spin-coated and vapor-deposited organic HTMs as well as insulating

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PMMA. All films were processed at room temperature without annealing. Measured permeability constants are plotted in Figure 2 as a function of HTM HOMO energy (data from refs

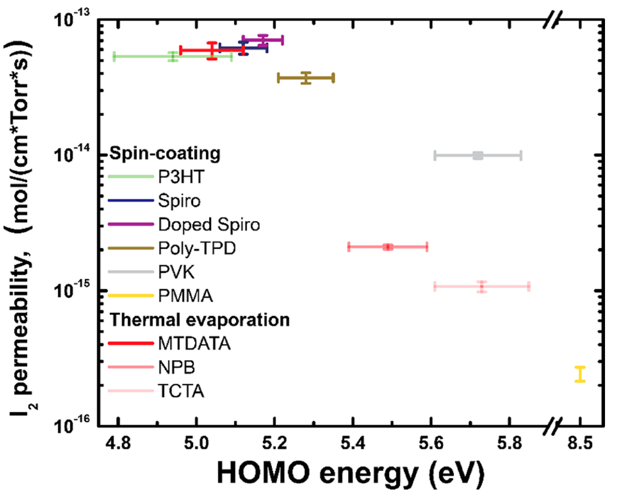

Figure 2. I_2 permeability constant of the indicated solution-processed and vacuum-deposited HTMs versus their HOMO energy at an I_2 partial pressure of approximately 0.2 Torr. Horizontal error bars reflect the standard deviation of (averaged) HOMO energies reported in the literature (refs 28−61), and vertical error bars are calculated from statistical variation measured across several samples.

28−61) ; values of P vary over nearly 3 orders of magnitude and are inversely correlated to the HOMO level. Notably, the trend appears to be independent of processing method (spin-coated versus vacuum-deposited) and molecular/polymeric structure (compare spiro-MeOTAD, poly-TPD, and PVK). However, while the HOMO energy determines the magnitude of P, processing details and physical properties also influence P. For instance, P of polymers was slightly decreased following densification by annealing near the glass transition temperature (SI, Figure S2). Importantly, the color change of the poly-TPD film in Figure 1b is indicative of a reaction simultaneously creating triiodide ions (iodide will become triiodide in the presence of excess I_2) and [poly-TPD]$^+$ polarons; HTM oxidation was confirmed by UV−vis absorbance measurements of spiro-MeOTAD films exposed to I_2 (SI, Figure S5).

HTM Oxidation/Doping by I_2. It is well-known that organic semiconductors can be oxidized (p-doped) by a reaction with I_2 vapor.[19−22] By simply omitting the Ag resistor in our Ag:I_2 test geometry, we fabricated lateral devices (20 μm channel) of as-deposited, undoped HTMs between ITO electrodes (Figure 3a) to monitor conductance changes during oxidation/doping reactions with I_2 vapor (see SI, Figure S6, for representative current transients). The change in electrical conductance after I_2 vapor exposure is plotted versus HOMO energy in Figure 3b. The magnitude of conductance change (and, indirectly, doping concentration) depends strongly on the HOMO energy; shallow HOMO energy materials are more strongly oxidized by I_2 (we cannot directly extract I_2 doping concentration as carrier mobility in organic semiconductors is strongly affected by Poole−Frenkel and trap-charge-limited transport effects).[62−64] This observation rationalizes the relationship between HOMO and permeability constant in Figure 2a; a strong reaction likely induces a larger concentration gradient and P (for now we assume that D is similar in magnitude for various HTMs).

Single- and multiple-electron reduction/oxidation mechanisms of iodine/diiodide/triiodide/iodide are numerous and complex[65] (see SI, Figure S7), and the exact potentials will additionally depend on the solvation environment in the organic solid. Therefore, we cannot determine the kinetic mechanism that determines the potential/HOMO energy threshold for a reaction with I_2, but we should note that the primary iodine speciation in the HTM is most likely triiodide, as reported by Koizumi et al. for poly(3-octylthiophene) doped by I_2 at a ratio of I_2 molecules:monomer units ≈ 1:37.[66] I_2 vapor doping experiments indicate that only a very deep HOMO > 5.4 eV—deeper than nearly all diiodide radical anion ($I_2/I_2^{-\bullet}$) and triiodide (I_2/I_3^-) redox potentials—will prevent all possible reactions with I_2 vapor, minimize I_2 uptake by the HTM, and decrease I_2 permeation rates.

Iodine Species Sourced from Perovskite. Our Ag:I_2 test critically reveals the iodine permeability dependence on HOMO energy when I_2 vapor was the reactant. We extend our study to confirm that the same trends are observed in a

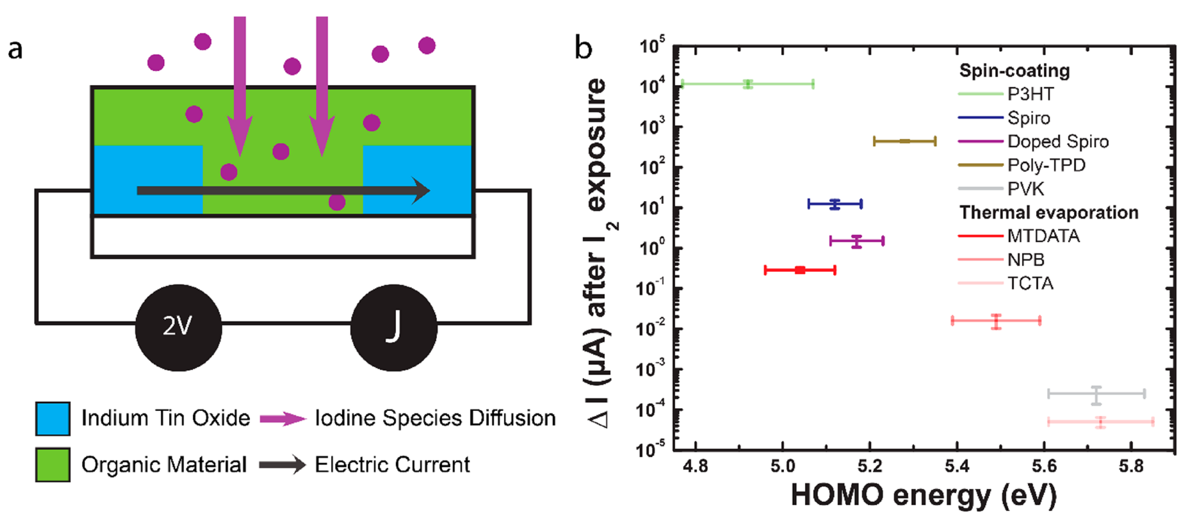

Figure 3. (a) Schematic of measurement configuration for characterizing doping of HTMs by I_2 vapor. (b) Change in current (final − initial) of the as-deposited HTMs following exposing to I_2 vapor.

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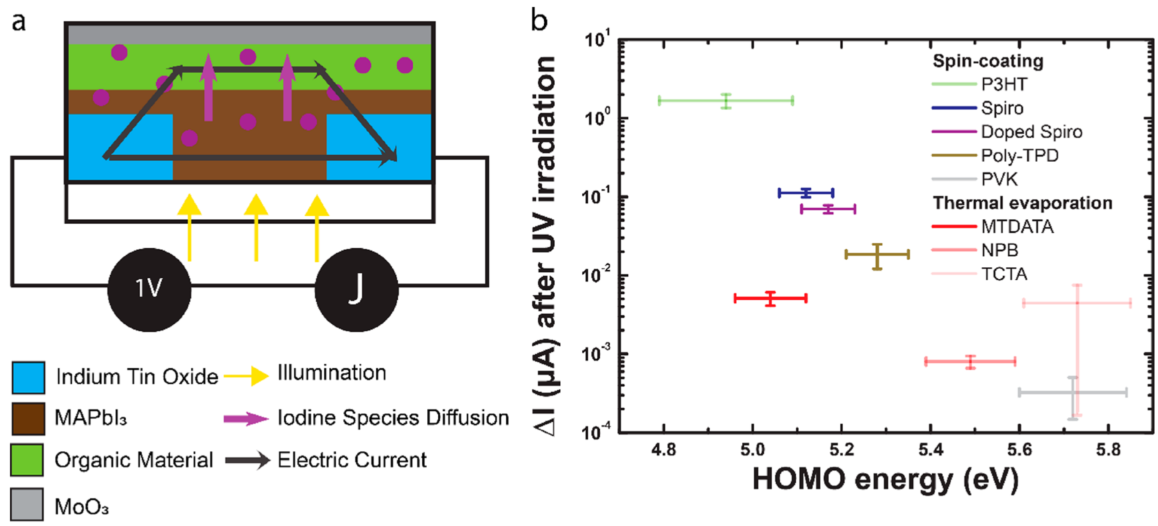

Figure 4. (a) Device structure and measurement scheme where the iodine source is a $MAPbI_3$ film encapsulated by the HTM. The current of the bilayer device is measured before and after UV illumination for 40 min from the $MAPbI_3$ side. (b) Absolute change in current (final measured 100 s after UV turned off) after UV illumination.

solid-state device where iodine-containing species are released during degradation of the halide perovskite active layer. Molecular I_2 is a proposed degradation product; moreover, I^0 radicals and HI are likely to be released prior to or simultaneously with I_2, although it should be noted that HI can readily decompose to I_2. We hypothesize that a strong doping reaction between I_2 vapor and an HTM may exacerbate reactions at the perovskite/HTM interface, resulting in larger amounts of triiodide uptake by the HTM (conversely, iodide loss from the perovskite) and faster permeation of triiodide throughout the device.

Current across lateral devices consisting of $MAPbI_3$/HTM/MoO_3 thin-film stacks (Figure 4a; MoO_3 served to prevent loss of I_2 to the environment; see SI, Figures S3 and S8) was measured before, during, and after 40 min of UV illumination to induce the release of photolysis products. In this device configuration, many HTM choices resulted in a large increase in conductance that persisted for long times (SI, Figures S9 and S10), which was not observed for control devices ($MAPbI_3$ and HTM-only lateral devices, SI, Figure S11). We mainly attribute the conductance increase to volatile HI and iodine-containing species photolyzed from the $MAPbI_3$ that quasi-irreversibly dope the HTM, similar to what has been induced electrochemically.[67]

Changes in absolute current after UV irradiation are plotted versus HOMO of the HTM in Figure 4b. Note that the change in absolute current is similar for undoped and doped spiro-MeOTAD, showing that the total amount of products reacted was the same in both cases and, thus, irrespective of initial doping. This measurement reveals that the relationship between HOMO level and doping by iodine species is maintained when the iodide originates from an interface with perovskite. Thus, shallow HOMO HTMs like poly(3-hexylthiophene) (P3HT) and spiro-MeOTAD are more likely to facilitate loss of iodide from the perovskite, effectively transport iodine throughout the device, and affect performance. The HTM is simply doped, which may not be directly detrimental (the opposite may even be true), but accelerating perovskite decomposition and metal electrode corrosion pathways is expected to negatively impact device stability.

In Figure 5, we use X-ray photoemission spectroscopy (XPS) to prove that perovskite degradation products contain iodine and confirm the transmission rate of such products depends on the HTM HOMO energy. Layered structures shown in Figure 5a, consisting of fluorine-doped tin oxide (FTO)/$MAPbI_3$/poly-TPD or PVK (~300 nm thick, unannealed)/Au (1 nm), were characterized by XPS, where the X-ray irradiation induced *in situ* degradation. The thin, likely discontinuous Au served as an indicator, as it irreversibly adsorbs iodine (without this indicator, iodine was lost to vacuum faster than it was generated and could not be detected). Figure 5b,c shows the I 3d XPS spectra for poly-TPD (HOMO ≈ 5.3 eV) and PVK (HOMO ≈ 5.7 eV) samples. These materials are both polymeric, have high T_g (>180 °C), and display vastly different I_2 vapor transmission rates (Figures 1b and 2). Initially, the I 3d XPS signal is nearly zero for both samples. However, during X-ray irradiation, measurable amounts of iodine migrate through the poly-TPD to the Au. In contrast, the PVK, having a HOMO energy deeper than the empirical 5.4 eV threshold, effectively blocks iodine transmission on this time scale (Figure 5c). These results clearly demonstrate that iodine species can be easily lost from the perovskite and permeate the bulk of an HTM if the HOMO is <5.4 eV.

Implications for Devices. Performance and current−voltage characteristics of perovskite devices have been proposed to be related to HTM HOMO energy and its proximity to various iodine reduction potentials.[24,68] Our combined observations support and explain recent reports that halide species permeate the HTM bulk in appreciable concentrations.[14,16,18−21,66] Thus, electrochemical features and current−voltage hysteresis are not limited to ion accumulation at perovskite/HTM interfaces, and the processes may be more complex than initially proposed. Our results also indicate that HTM dedoping by iodide may be insignificant in solid-state perovskite devices where I_2 released from the perovskite instead contributes to further doping (Figure 4b).[13,67] Ultimately, the ideal HTM should have as deep a HOMO as possible while still efficiently extracting holes from the perovskite, a strategy that proved successful for Hou et al.[68] Long-term, I_2 permeation through organic HTMs may be

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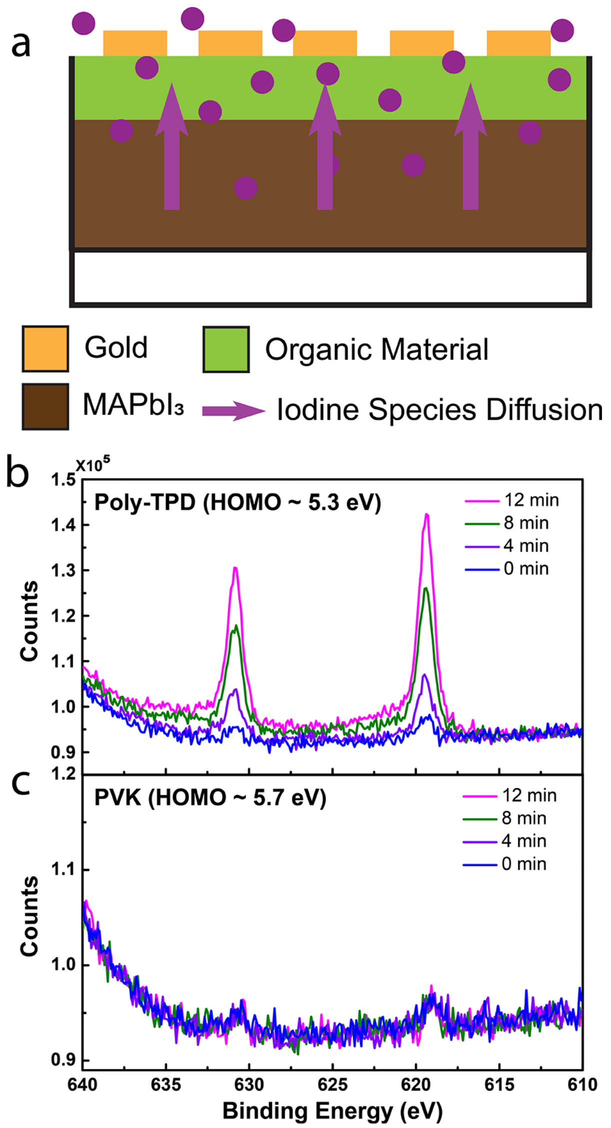

Figure 5. (a) Schematic of sample used to measure loss of iodine from $MAPbI_3$ and subsequent transmission through an HTM to adsorb to Au. I 3d XPS measurements of (b) FTO/$MAPbI_3$/poly-TPD/Au and (c) FTO/$MAPbI_3$/PVK/Au samples over time under X-ray irradiation.

inevitable in solar cells, though, since deep HOMO energy HTMs will not efficiently collect current due to a barrier for hole extraction. Buffer layers such as metal oxides, like we show for MoO_3, may effectively protect metal electrodes from iodine-saturated HTMs.[10,12,42,69]

Last, we recognize that I_2 or I_3^- can induce C−H or C−C bond-breaking/forming chemical reactions in organic molecules and polymers. Poly(3-alkylthiophenes) doped by I_2 are known to deprotonate at the α-C, leading to dedoping, HI release, and cross-linking.[66] Indeed, following extended exposure to I_2 vapor (12 h) and removal of the I_2, P3HT films became insoluble in chlorobenzene, suggesting cross-linking occurred (SI, Figure S12). Poly(dimethylsiloxane) (PDMS) also reacts chemically with I_2, becoming brittle. These observations exemplify the capability of iodine species within organic materials to drive more harmful chemistry than charge-transfer and doping reactions. Thus, the combination of Ag:I_2 corrosion and I_2 vapor doping tests provides opportunities to reveal chemical impacts of iodine on organic hole and electron transport materials and encapsulants used in perovskite devices, lending greater insight into potential degradation mechanisms. Further investigation of perovskite/HTM interfaces is needed to fully elucidate the interplay between molecular structure, energetics, density, polarity, reduction/oxidation, and potential electrochemical reactions dictating the performance and stability of perovskite devices.

In summary, we have developed a modified Ag:I_2 corrosion test modeled after the Ca:O_2 test to characterize the I_2 permeability of common organic HTMs. We find that transmission rate inversely correlates to the ionization/HOMO energy of the organic material. High I_2 permeability is additionally related to the ability of I_2 or I_2-like species to react with and dope shallow HOMO energy HTMs. The HOMO energy dependence was maintained in solid-state perovskite structures where $MAPbI_3$ degradation products induced by UV or X-ray irradiation were the only source of iodine. These insights firmly establish the influence of HTM HOMO energy on the rate of iodide diffusion in halide perovskite devices. As such, we introduce an additional design rule for HTMs—beyond pinholes and grain boundaries—to optimize device performance *and stability* by careful consideration of the HOMO energy to impede degradation pathways associated with halide migration.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsenergylett.0c02495.

> Description of experimental methods, tabulated HTM HOMO energies and molecular structures, discussion of diffusion model and computations, additional Ag/HTM and control device current versus time measurements under various conditions, UV−vis−NIR of spiro-MeOTAD exposed to I_2, iodide/diiodide/triiodide reaction mechanisms and potentials compared to HTM HOMOs used in this study, and photographs of P3HT reacted with I_2 vapor and "dissolved" in chlorobenzene (PDF)

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Notes

The authors declare no competing financial interest.

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